Day 13 | Thursday

13

November 19, 2026

Ratios, rates and proportions

12:45 PM–2:00 PM • Math • 75 min

Scale drawings and weekly check

2:15 PM–3:30 PM • Math • 75 min

Week 4 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Ratios, rates and proportions

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 4, Lesson 7

2:15 PM–3:30 PM • 75 min

Scale drawings and weekly check

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 4, Lesson 8

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Ratios, rates and proportions

Today’s goal: Simplify ratios in common units and use proportions to scale a material quantity.

Share a prior use of this skill; identify the quantity and unit.

Textbook sections 6.5–6.7 pp.98–101

Exercise Book 6A Q14–18 (p.40); 6B Q4,6 (p.41), Q7 (p.42)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 6.5–6.7 pp.98–101

Recall and reconnect

Use Exercise Book 6A Q14. Explain why millimetres and metres must match before comparing quantities.

Attempt the short retrieval task independently; explain the method before seeing answers.

Resources: Exercise Book 6A Q14–18 (p.40); 6B Q4,6 (p.41), Q7 (p.42)

Textbook sections 6.5–6.7 pp.98–101

Recall from the Exercise Book | 6A Q14

Write 36 : 84 in lowest terms.

Write the operation, working and units before checking.

Textbook sections 6.5–6.7 pp.98–101

Write ratios with clear units

1. A ratio compares quantities in a stated order.

2. Convert comparable lengths to the same unit before simplifying.

3. Divide both terms by the same factor to keep the ratio equivalent.

4. A rate retains its units, such as dollars per metre.

Textbook sections 6.5–6.7 pp.98–101

Scale every part by the same factor

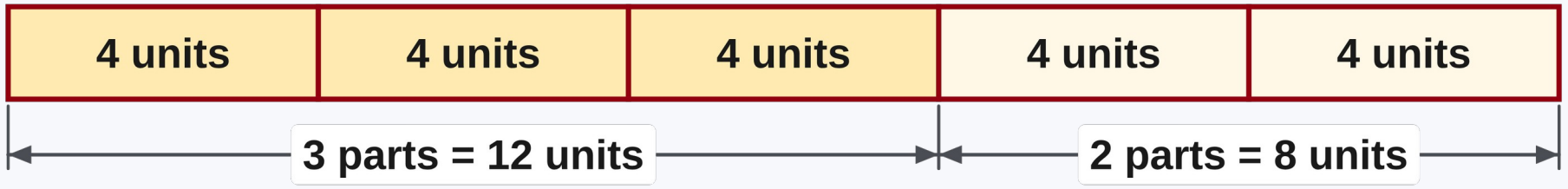
1 Add the ratio terms to find the number of equal parts.

2 Find the value of one part, then multiply for each required share.

3 For a proportion, keep corresponding quantities in the same order.

4 Check the scaled quantities against the original ratio.

Ratio 3 : 2 of 20 units



Textbook sections 6.5–6.7 pp.98–101

First make the comparison meaningful

2 m : 500 mm = 2,000 : 500 = 4 : 1

Classroom ratio 3:2, total 20 units:
5 total parts → 20 ÷ 5 = 4 units per part → 12 and 8 units.
A 3:2 ratio does not mean the first part is 3/2 of the whole.

2 m = 2,000 mm	**500 mm**	**500 mm**	**500 mm**	**500 mm**	**4 parts**
500 mm	**500 mm**				**1 part**

Together: 6A Q15,17–18 and 6B Q6. Label rate units.
Use proportional Shape Canvas rectangles; check with the calculator after writing the proportion.

Textbook | page 99

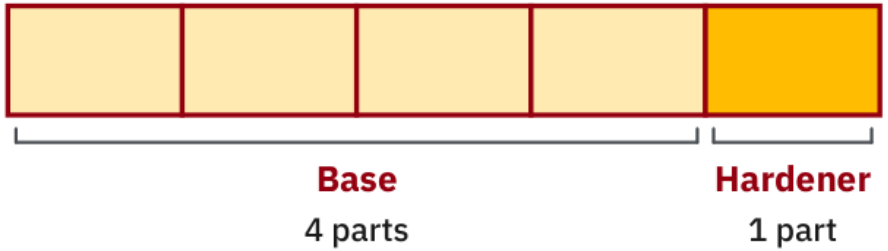


Figure 6.2 A mix of 4 : 1. Every part is the same size.

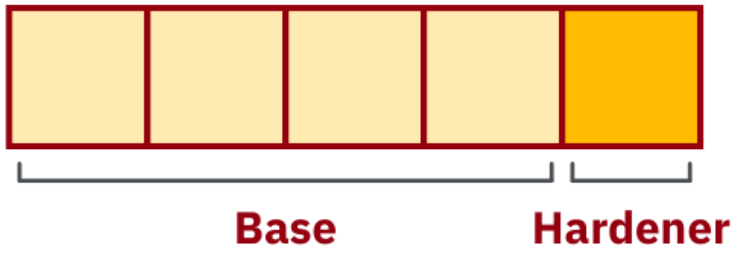
Textbook p.99 • Figure 6.2 — Equal parts in a 4:1 ratio

Textbook sections 6.5–6.7 pp.98–101

Textbook | page 99

Example 6.8 Sharing an amount in a ratio

You need 750 mL of mixed coating. The label gives a mix of 4 parts base to 1 part hardener. How much of each do you measure?



SOLUTION

1. Add the parts: 4 + 1 = 5 parts.
2. One part: 750 ÷ 5 = 150 mL.
3. Base: 4 × 150 = 600 mL. Hardener: 1 × 150 = 150 mL.
4. Check: 600 + 150 = 750 mL.

Textbook p.99 • Example 6.8 — Sharing a total in a ratio; classroom math example

Textbook sections 6.5–6.7 pp.98–101

Spot the mistake. Explain the correction.

Can you simplify 1 m : 50 mm as 1 : 50?

Discuss the method before checking the answer.

No. Convert to common units first: 1,000 mm : 50 mm, then simplify both terms.

Textbook sections 6.5–6.7 pp.98–101

Work together

Work Exercise Book 6A Q15,17–18 and 6B Q6 in pairs. Label numerator and denominator units.

Work in pairs; alternate calculator operator and checker. Show units and an estimated answer.

Resources: Exercise Book 6A Q14–18 (p.40); 6B Q4,6 (p.41), Q7 (p.42)

Textbook sections 6.5–6.7 pp.98–101

Guided practice | 6A Q15

Write the ratio of 250 mm to 1.5 m in lowest terms.

Work with a partner. Take turns explaining the next step.

Textbook sections 6.5–6.7 pp.98–101

Compare your method | 6A Q15

Write the ratio of 250 mm to 1.5 m in lowest terms.

CHECK TOGETHER

1 : 6

250 : 1,500, divide both by 250

Textbook sections 6.5–6.7 pp.98–101

Your scheduled practice

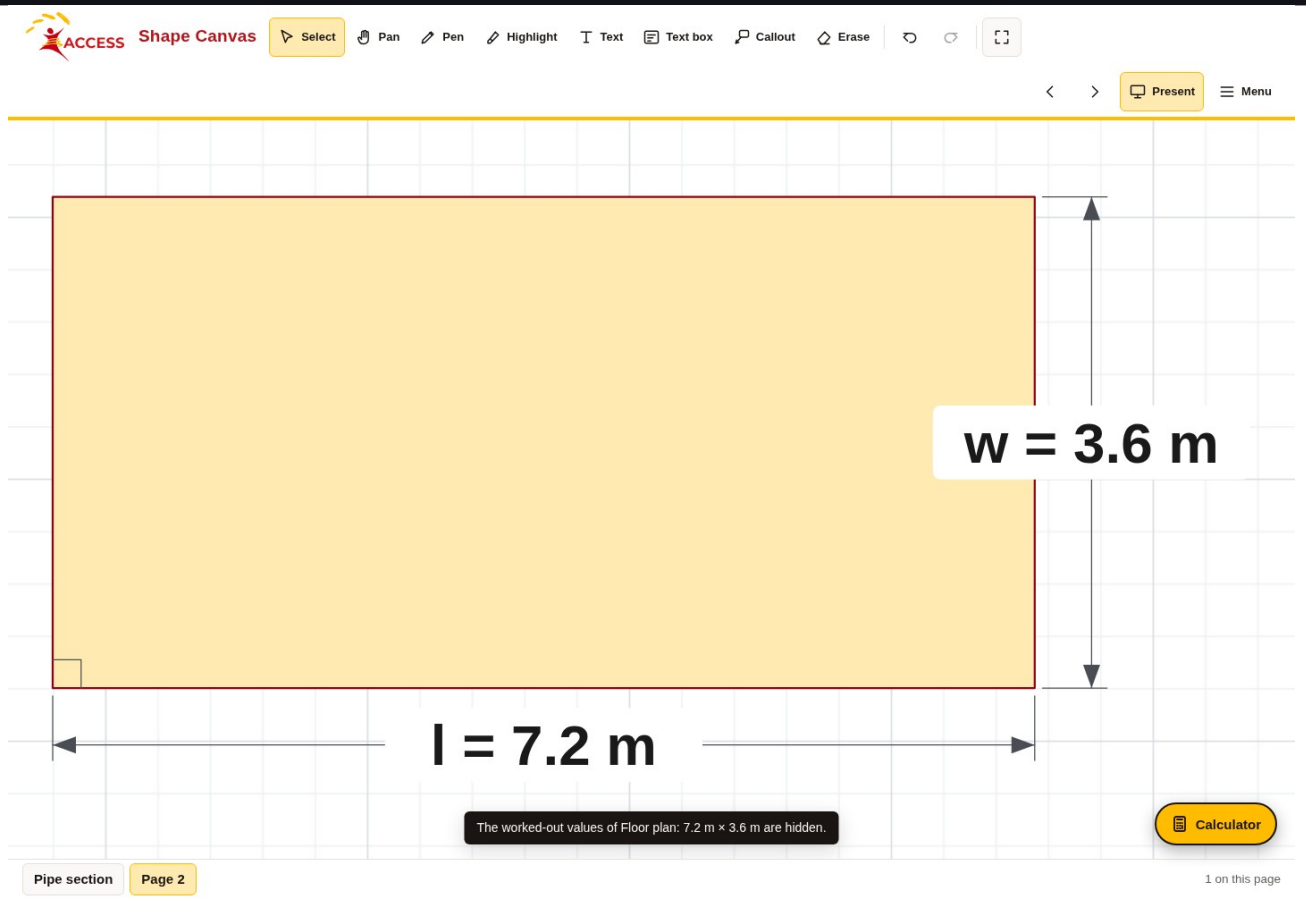
Complete Exercise Book 6B Q4 and Q7. Draw proportional parts on Shape Canvas; treat the book’s epoxy mix only as a math example, not a real mixing instruction.

Solve on paper first; use the tool to check; correct one error with a written reason.

Resources: Exercise Book 6A Q14–18 (p.40); 6B Q4,6 (p.41), Q7 (p.42)

Textbook sections 6.5–6.7 pp.98–101

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 6.5–6.7 pp.98–101

Practise in Numeracy Hub and Shape Canvas

1. Write the assigned ratio or proportion with units and solve it on paper.

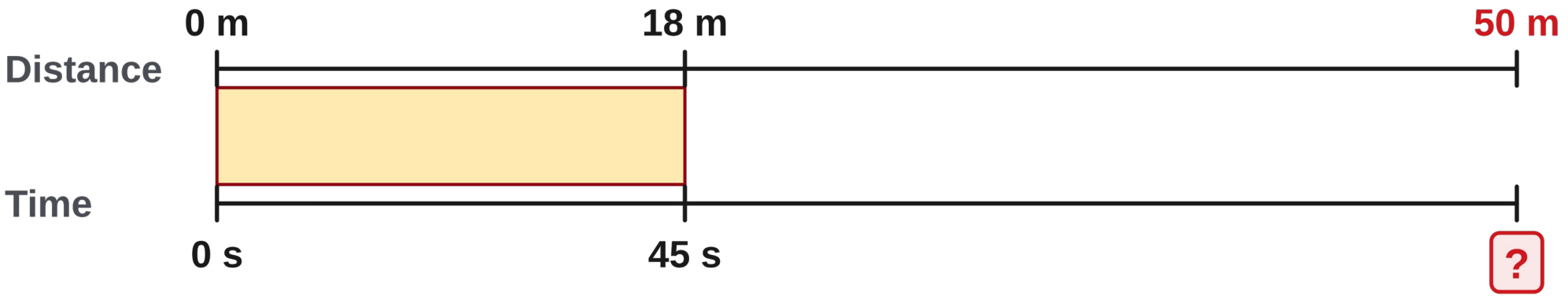
2. Compare Shape Canvas rectangles whose dimensions use the same scale factor.

3. Check the rate or proportion with Numeracy Hub Calculator and explain the units.

Textbook sections 6.5–6.7 pp.98–101

Practise from the Exercise Book | 6B Q4

A conveyor belt moves 18 m in 45 s. What is its rate in m per s? At that rate, how long does a part take to travel 50 m?

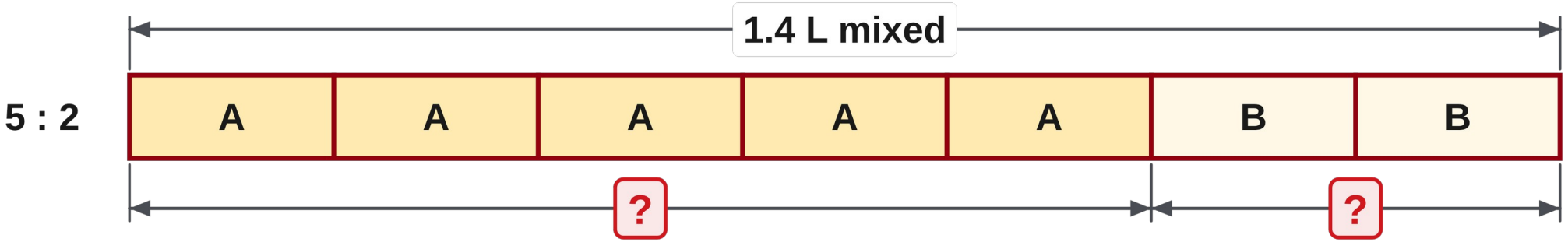


Write the operation, working and units before checking.

Textbook sections 6.5–6.7 pp.98–101

Practise from the Exercise Book | 6B Q7

A two-part epoxy is mixed 5 : 2 by volume. You need 1.4 L of mixed epoxy. How much of each part do you measure?



Write the operation, working and units before checking.

Textbook sections 6.5–6.7 pp.98–101

Check independently

Exercise Book 6A Q18 and Exercise Book 6B Q6; show the proportion or unit-rate calculation.

Solve without displayed answers; include working, units and the requested rounding.

Resources: Exercise Book 6A Q14–18 (p.40); 6B Q4,6 (p.41), Q7 (p.42)

Textbook sections 6.5–6.7 pp.98–101

Independent check | 6A Q18

Solve $\frac{7}{x} = \frac{21}{36}$.

Answer independently. Keep worked solutions closed.

Textbook sections 6.5–6.7 pp.98–101

Independent check | 6B Q6

Cutting 14 brackets uses 9.8 m of angle iron. How much angle iron do 35 brackets use?

Answer independently. Keep worked solutions closed.

Textbook sections 6.5–6.7 pp.98–101

Wrap up and choose the next step

Ask: What did you learn? Where will you use it? What will you check next time?

Name the math and one other essential skill used; write the next practice step.

Resources: Exercise Book 6A Q14–18 (p.40); 6B Q4,6 (p.41), Q7 (p.42)

Textbook sections 6.5–6.7 pp.98–101

Break

2:00 PM–2:15 PM

Return ready for: Scale drawings and weekly check

This break is excluded from teaching minutes.

Scale drawings and weekly check

Today’s goal: Translate drawing lengths to real lengths and explain a multi-step calculation from a dimensioned sketch.

Share a prior use of this skill; identify the quantity and unit.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Exercise Book 6A Q19–20 (p.40); 6B Q8 (p.42); 6C Q5,7–8 (p.44)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Recall and reconnect

Identify what 1:40 means. Compare a drawing length and a real length before calculating.

Attempt the short retrieval task independently; explain the method before seeing answers.

Resources: Exercise Book 6A Q19–20 (p.40); 6B Q8 (p.42); 6C Q5,7–8 (p.44)

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Read the direction of a scale

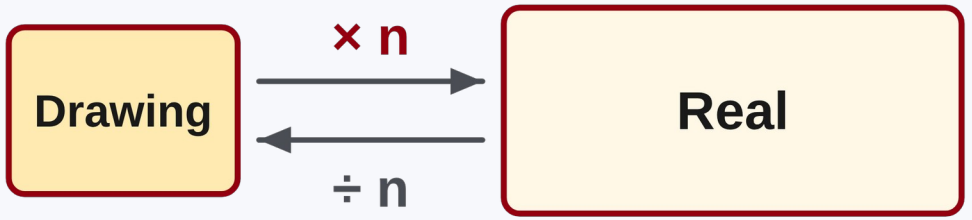
1 A 1:40 scale means one drawing unit represents 40 matching real units.

2 For 1:n reductions: drawing → real, multiply by n; real → drawing, divide by n.

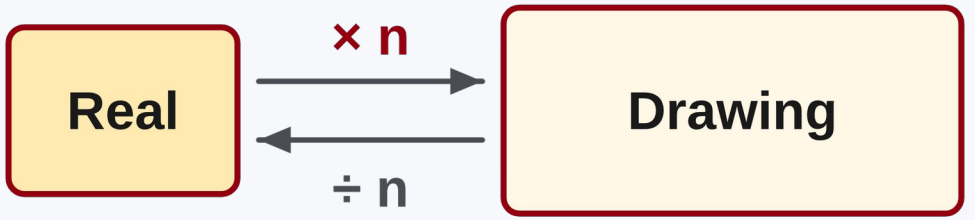
3 For n:1 enlargements: drawing → real, divide by n; real → drawing, multiply by n.

4 Convert units when needed; keep drawing and real dimensions labelled.

1 : n reduction



n : 1 enlargement



Textbook sections 6.8 pp.102–103; review Chapter 4–6

Use dimensions, not screen size

1. Read the written measurement on a drawing.
2. A resized screen image is not a reliable ruler.
3. Write a sketch and an equation showing which length is unknown.
4. Check direction: a 1:40 drawing is smaller; a 3:1 drawing is larger than the real object.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Multiply to return to full size

6A Q19: 85 mm × 40 = 3,400 mm = 3.4 m

Reverse check: 3,400 mm ÷ 40 = 85 mm.
Classroom example: 2,000 mm at 1:50 is drawn as 40 mm.



Together: 6A Q20 and 6B Q8. Explain where inches, feet and the scale factor enter. Do unit conversions explicitly.

Shape Canvas: label a rectangle, then change its dimensions proportionally. Zooming alone does not change the labelled dimensions.

Textbook | page 102

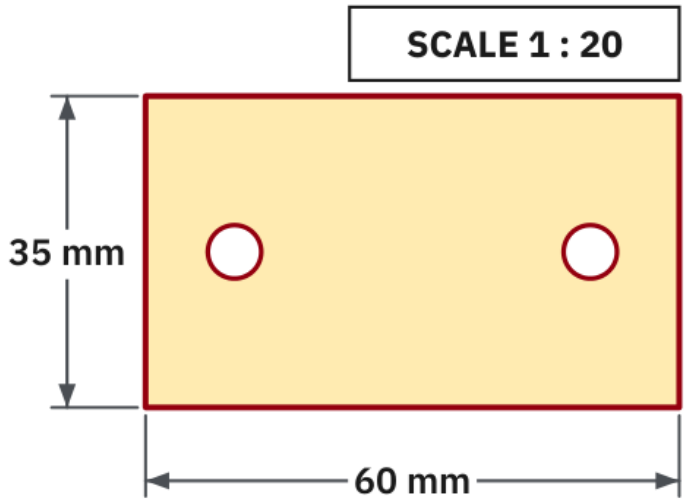


Figure 6.3 A plate drawn at 1 : 20. The sizes shown are the lengths measured on the paper.

The plate above measures 60 mm by 35 mm on the paper, so the real plate is 60 × 20 = 1,200 mm long and 35 × 20 = 700 mm wide.

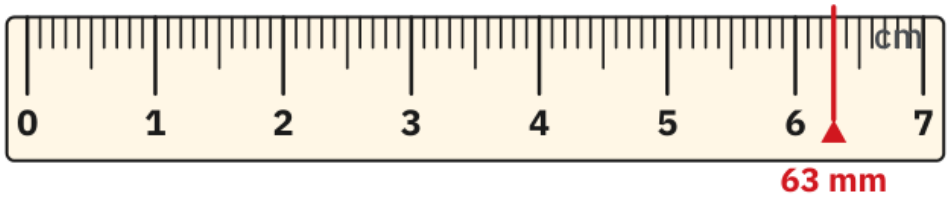
Textbook p.102 • Figure 6.3 — Drawing dimensions and real dimensions

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Textbook | page 102

Example 6.11 Measuring a drawing

A line on a drawing at 1 : 50 measures 63 mm with a rule, as shown. How long is the real part?



SOLUTION

1. Drawing to real, so multiply by 50.
2. 63 × 50 = 3,150 mm.

Textbook p.102 • Example 6.11 — Scale factor from drawing to real length

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Spot the mistake. Explain the correction.

A drawing is reduced to 1:40. Should the real length be divided by 40 again?

Discuss the method before checking the answer.

From drawing to real, multiply by 40. Divide only when going from real to drawing.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Work together

Work Exercise Book 6A Q20 and 6B Q8 with a partner. Explain where inches, feet and the scale factor enter.

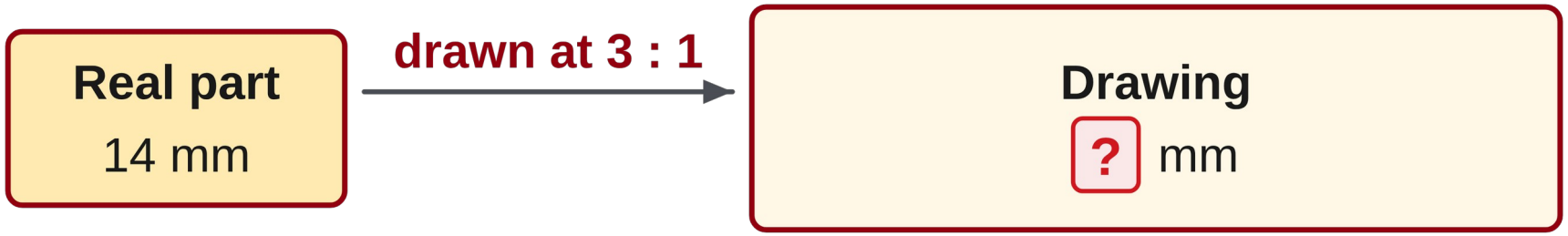
Work in pairs; alternate calculator operator and checker. Show units and an estimated answer.

Resources: Exercise Book 6A Q19–20 (p.40); 6B Q8 (p.42); 6C Q5,7–8 (p.44)

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Guided practice | 6A Q20

A 14 mm part is drawn at 3 : 1. How long is it on the drawing?



Work with a partner. Take turns explaining the next step.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Compare your method | 6A Q20

A 14 mm part is drawn at 3 : 1. How long is it on the drawing?

CHECK TOGETHER

42 mm

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Your scheduled practice

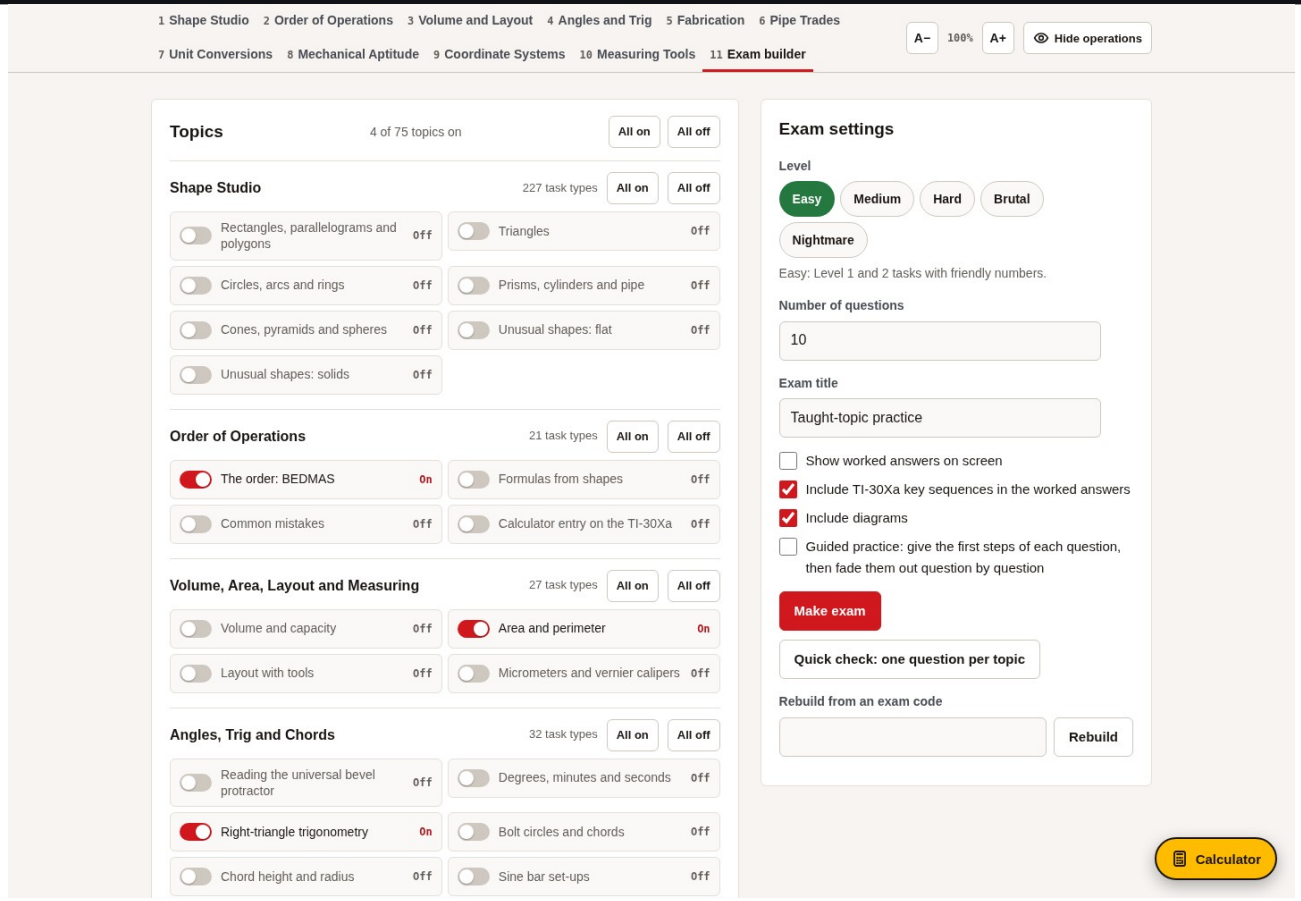
Complete Exercise Book 6C Q5,7–8. Use a four-question paper checkpoint from the week; optional Numeracy Hub review only after core work.

Solve on paper first; use the tool to check; correct one error with a written reason.

Resources: Exercise Book 6A Q19–20 (p.40); 6B Q8 (p.42); 6C Q5,7–8 (p.44)

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Practise in Numeracy Hub and Shape Canvas

1. Build the assigned dimensioned rectangle in Shape Canvas from written sizes.

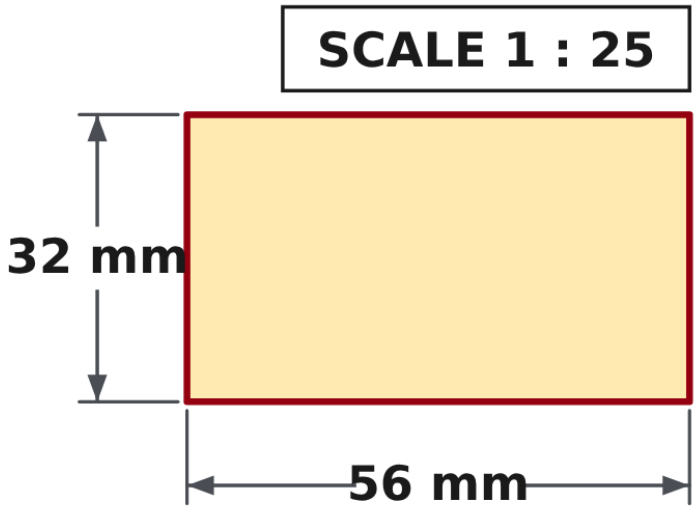
2. Change one dimension by the stated factor and predict the resulting labelled dimension.

3. Complete the core paper work first; use optional Easy Exam builder review on taught topics only.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Practise from the Exercise Book | 6C Q5

The sizes are measured on a 1 : 25 drawing. Find the real length and width.

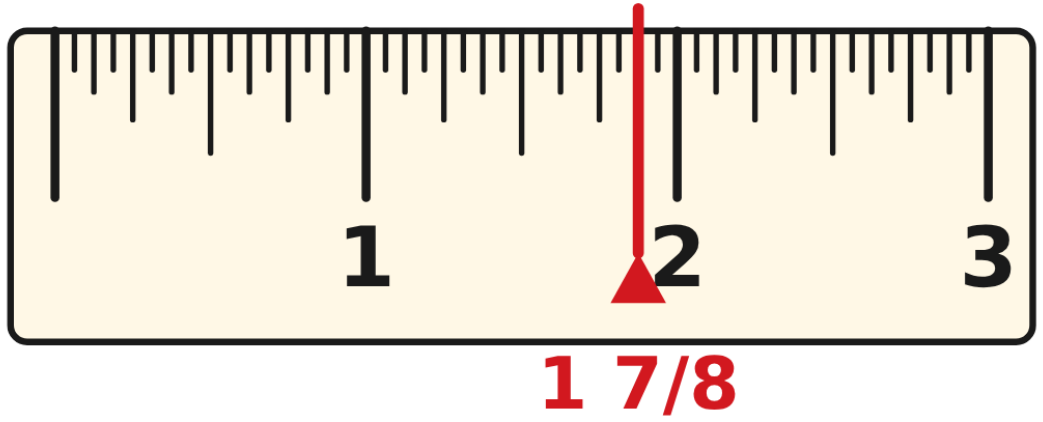


Write the operation, working and units before checking.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Practise from the Exercise Book | 6C Q7

A drawing is marked 1/2″ = 1′-0″. Find the real length of the line in feet and inches.



Write the operation, working and units before checking.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Check independently

Exercise Book 6A Q19–20 independently, with units and a sketch showing scale direction.

Solve without displayed answers; include working, units and the requested rounding.

Resources: Exercise Book 6A Q19–20 (p.40); 6B Q8 (p.42); 6C Q5,7–8 (p.44)

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Independent check | 6A Q19

At a scale of 1 : 40, a line measures 85 mm. How long is the real length?

Answer independently. Keep worked solutions closed.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Independent check | 6A Q20

A 14 mm part is drawn at 3 : 1. How long is it on the drawing?

Answer independently. Keep worked solutions closed.

Textbook sections 6.8 pp.102–103; review Chapter 4–6

Wrap up and choose the next step

Ask: What did you learn? Where will you use it? What will you check next time?

Name the math and one other essential skill used; write the next practice step.

Resources: Exercise Book 6A Q19–20 (p.40); 6B Q8 (p.42); 6C Q5,7–8 (p.44)

Textbook sections 6.8 pp.102–103; review Chapter 4–6